

1. Project Title

Automated Water Filtration Design and Recommendation System

2. Problem Statement

Water quality varies widely depending on geography, source, and environmental exposure. Industrial facilities, housing complexes, schools, hospitals, and residential homes increasingly rely on engineered filtration systems to ensure safe and usable water. Designing these filtration systems correctly is critical, because an incorrect design can directly affect health, equipment performance, and operational costs.

Today, most filtration design processes are highly manual. Engineers study the laboratory report, interpret chemical values, choose appropriate treatment technologies, calculate vessel capacities, determine media quantities, and finally prepare diagrams and bills of materials. Each of these steps is prone to human interpretation and fatigue-based errors. Junior engineers or technicians may misread thresholds, over-size or under-size vessels, or choose inappropriate filtration media.

As the number of installations grows, organizations require faster turnaround, better accuracy, and repeatable design logic. Without automation, engineers spend unnecessary time repeating the same calculations and technical checks.

At the same time, customers expect transparency: they want to understand why a specific filtration system was chosen. Manual explanations often lack consistency.

Therefore, the problem is not the availability of filtration technologies — it is the **lack of a smart, automated, standardized design tool** that converts raw water test reports into an accurate and structured filter design.

The proposed project addresses this gap by developing a system that processes water quality data, applies engineering logic, and produces reliable filtration recommendations automatically .

3. Objectives of the Project

The project is designed with the following detailed objectives:

Automate report interpretation

Create a digital system capable of reading water quality values through OCR or manual entry and interpreting them without human bias.

Support standardized engineering logic

Implement industry-accepted threshold rules, formulas, and design calculations to avoid guesswork.

Generate accurate filtration recommendations

Choose suitable filtration media based on water characteristics such as TDS, turbidity, hardness, chloride, iron, and hydrogen sulfide.

Perform automated vessel and media calculations

Compute vessel size, freeboard, media quantity, and layering based on flow rate and required contact time.

Improve technical communication

Produce diagrams, BOM lists, and structured reports that both technicians and customers can understand.

Reduce dependency on expert designers

Allow junior engineers and technicians to confidently generate reliable system designs.

Save time and reduce repetitive work

Replace spreadsheet-based calculations and manual drawings with automated reporting.

Collectively, these objectives aim to improve design transparency, decision accuracy, and system reliability .

4. System Overview

The proposed system is a **smart design assistant** for water treatment professionals. The application accepts data from two primary sources:

OCR Input – Users upload a JPG test report. Using Optical Character Recognition, the system detects parameter names and extracts numerical values.

Manual Input Form – Users can type values directly, update incorrect OCR readings, or enter new test results.

Once the values are available, the **Design Engine** evaluates each parameter using rules and threshold logic. For example, if iron is detected above permissible limits, oxidation media such as Manganese Dioxide or Birm may be recommended. If TDS values exceed standards, the system may recommend RO or resin conditioning.

The engine then calculates recommended vessel dimensions based on flow rate, vessel geometry, contact time, and freeboard space. It automatically estimates how much media of each type is required and the appropriate order in which layers must be arranged.

The application outputs:

a Bill of Materials, a visual cross-sectional diagram of the vessel, and a printable report ready for client communication.

This approach reduces human intervention while keeping the final decision transparent and editable. It bridges engineering expertise with usability by providing clarity for both sales teams and technical staff .

5. Key Features / Modules

1. Optical Character Recognition (OCR) Module

Extracts data automatically from scanned or photographed lab reports. Enhances contrast, searches keywords, and collects corresponding values.

2. Manual Data Entry Module

A structured form that allows cleaning, editing, or replacing values.

3. Validation & Review Layer

Displays extracted results in editable tables for final confirmation before design processing.

4. Design Logic Module

- Applies threshold-based rules. For example:
- Iron > 0.3 mg/L → Manganese Dioxide / Birm
- TDS > 500 ppm → RO or resin-based system
- H₂S detected → Catalytic or Activated Carbon
- High turbidity → Multi-grade sand or anthracite

5. Vessel Calculation Module

Computes total media volume, ensures freeboard, and maps values to standard vessel sizes.

6. Layering & Structural Module

Arranges gravel, sand, carbon, resin, and specialty media according to density and hydraulic principles.

7. BOM Generator

Produces detailed material lists including media name, volume, and vessel specifications.

8. Visual Representation Module

Creates clear, color-coded diagrams representing internal structure and material placement.

9. PDF Report Generator

Compiles results into a downloadable and printable report for customers and installation teams.

All these modules collectively ensure automation, clarity, and speed in filtration design .

6. Methodology / Technologies Used

The development follows a structured approach:

Phase 1: Requirement Analysis

Understanding water parameters, design standards, and existing manual workflows.

Phase 2: System Design

Creating architecture diagrams, database structure, module layout, and interface flow.

Phase 3: Implementation

Building each module in small iterations and testing continuously.

Phase 4: Integration

Combining OCR, design engine, calculations, and reporting modules into one system.

Phase 5: Testing & Validation

Checking:

OCR accuracy, correctness of formulas, rendering speed, usability of reports.

Phase 6: Deployment

Preparing the application for real-time use and feedback collection.

Technology Stack (Proposed):

- Frontend: HTML / CSS / JS or React
- Backend: Python / Node.js / Java
- Database: MySQL / MongoDB
- OCR: Tesseract OCR
- Graphics: SVG / Canvas
- Report Engine: PDF export libraries

This ensures maintainability, scalability, and performance across environments.

7. Scope of the Project

In-Scope

- OCR reading of water reports
- Entry and validation of chemical values
- Rule-based recommendation engine
- Vessel volume calculation
- Layering visualization
- BOM and report generation

Out-of-Scope (Phase-1)

- Real-time IoT water monitoring
- Automatic costing and quotations
- Mobile apps
- AI-driven prediction or analytics

By defining a focused but impactful scope, the project remains realistic and achievable while building a strong foundation for future expansions.

8. Expected Outcomes

Upon completion, the project is expected to deliver:

- A functional system that transforms water quality data into structured designs.
- Faster project turnaround for engineers and sales teams.
- Reduced dependency on manual calculations and notebooks.
- Increased reliability, accuracy, and accountability of filtration system recommendations.
- A visual, educational representation that helps installers assemble systems correctly.

- A reusable framework that can later integrate costing, quotation generation, and IoT-based monitoring.
- The final product supports engineering clarity and professional workflow improvement, aligning perfectly with the goals expressed in the SRS .